Classical Parameter Estimation & Quant Puzzles — Complete Formula Reference

Every identity from Ch. 2.1, zero derivations, plus the extra results your 20-problem set actually used.

1. Maximum Likelihood Estimation (MLE)

- **Procedure:** maximize $L(\theta) = \prod_i f(x_i;\theta)$, usually via $\frac{d}{d\theta}\ln L(\theta) = 0$
- **Binomial p:** $\hat{p} = k/n$ (sample proportion)
- **Poisson λ:** $\hat{\lambda} = \bar{x}$ (sample mean)
- **Geometric p (on $\{1,2,\dots\}$):** $\hat{p} = 1/\bar{x}$
- **Discrete Uniform$\{1,\dots,N\}$ (1 or more obs.):** $\hat{N}_{\text{MLE}} = \max(\text{data})$ — likelihood is $\propto \binom{N}{k}^{-1}$, strictly *decreasing* in N, so maximized at the smallest admissible value
- **Invariance property:** if $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$ for *any* function g — never re-derive from scratch for a transformed parameter
- **Sufficiency:** a statistic T is sufficient for θ if the likelihood ratio $L(\theta_1)/L(\theta_2)$ depends on the data only through T — e.g. k (fill count) is sufficient for Binomial p; the exact sequence adds nothing

trap: MLE for a boundary/support parameter (discrete Uniform's N) behaves completely differently from MLE for an interior parameter (p, λ) — it sits at the edge of what's logically possible rather than at a calculus-style turning point, and is systematically biased as a result (see Section 3).

2. Method of Moments (MoM)

- **One-parameter procedure:** set population mean = sample mean, solve for θ
- **Poisson:** $\hat{\lambda}_{\text{MoM}} = \bar{x}$ — identical to MLE
- **Discrete Uniform$\{1,\dots,N\}$:** $\mathbb{E}[X] = \frac{N+1}{2} = \bar{x} \implies \hat{N}_{\text{MoM}} = 2\bar{x} - 1$
- **Two-parameter procedure (e.g. Binomial(n,p), both unknown):** match mean *and* variance:

$$np = \bar{x}, \quad np(1-p) = s^2$$

$$\implies \hat{p} = 1 - \frac{s^2}{\bar{x}}, \quad \hat{n} = \frac{\bar{x}}{\hat{p}}$$

- **General rule:** match as many moments as you have unknown parameters, no more, no fewer

trap: MoM can produce a logically impossible estimate (e.g. $\hat{N}_{MoM} < \max(data)$ when an outlier is present) — unlike MLE, it has no built-in mechanism to respect hard structural constraints. Always sanity-check a MoM output against the data's logical bounds.

3. The German Tank Problem & the MVUE

- **Setup:** k serial numbers sampled without replacement from $\{1,\dots,N\}$; $m = \max(\text{sample})$
- **MLE is biased:** $\mathbb{E}[m] = \frac{k(N+1)}{k+1}$ — always *less* than N
- **The MVUE correction:**

$$\hat{N} = m + \frac{m}{k} - 1 = \frac{k+1}{k}\,m - 1$$

- **Unbiasedness (proof sketch):** substitute $\mathbb{E}[m]$ above into $\mathbb{E}[\hat{N}]$; the $\frac{k+1}{k}$ and $\frac{k}{k+1}$ factors cancel exactly, leaving $\mathbb{E}[\hat{N}] = N$
- **Variance of the MVUE:**

$$\text{Var}(\hat{N}) = \frac{(N-k)(N+1)}{k(k+2)}$$

(derived from $\text{Var}(m) = \frac{k(N-k)(N+1)}{(k+1)^2(k+2)}$ via the linear transform)

trap: "unbiased" ≠ "precise" — $\hat{N}$ has zero bias but can still have a large standard deviation with small k. Don't mistake an unbiasedness proof for a precision guarantee.

4. Bias, Variance & Mean Squared Error

- **Bias:** $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$
- **Core decomposition:** $\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$
- **Unbiased estimator:** MSE = Var exactly (bias term vanishes)
- **Shrinkage / regularized estimator pattern (e.g. Binomial):**

$$\tilde{p} = \frac{k+1}{n+2}$$

("add-one" / Laplace smoothing) trades a small bias (pulls toward 0.5) for a variance reduction (divides by $(n+2)^2$ instead of n^2) — can have *strictly lower* MSE than the unbiased MLE

- **MSE gap under more data:** both bias and variance shrink as sample size grows, but the *relative* MSE gap between a biased and an unbiased/corrected estimator does not necessarily close

trap: "unbiased" and "best" are not synonyms. MSE is the only honest single-number comparison between estimators — a biased estimator with sufficiently lower variance can beat an unbiased one on MSE every time (this is the entire logic behind ridge regression and Bayesian shrinkage priors).

Quick Decision Guide — "Which Tool Do I Need?"

- Asked for "the most likely value of θ given this data" → **MLE**: write $L(\theta)$, take ln, differentiate, solve (or recognize it's one of the standard results in Section 1).
- Parameter lives in the interior of its range (e.g. a probability $p \in (0,1)$, a rate $\lambda > 0$) → MLE comes from a calculus turning point; usually reduces to a simple sample statistic.
- Parameter is a **boundary/support** parameter (e.g. the upper limit N of a discrete Uniform) → MLE sits at the edge of the feasible region (= max of the data), not an interior optimum — and is predictably biased low.
- Need two unknown parameters from one distribution family → **two-moment Method of Moments**: match mean and variance, solve the two equations simultaneously.
- A MoM estimate looks suspicious (e.g. smaller than an observed data point that logically bounds it) → distrust it; check against MLE or a dedicated unbiased estimator instead.
- Estimating the size of a population/pool from a sample's **maximum** (serial numbers, locomotive numbers, user IDs) → this is the **German Tank Problem**: use the MVUE $\hat{N} = m + \frac{m}{k} - 1$, not the raw max.
- Comparing two or more candidate estimators → compute **MSE** for each (Bias^2 + Var), never compare bias or variance alone in isolation.
- A "biased" estimator is on the table (shrinkage, regularization, a Bayesian-flavored formula) → don't dismiss it on bias alone — compute its MSE; it may win.
- Need the MLE of some function of a parameter you've already estimated (odds, log-odds, a ratio) → **invariance**: just apply the function to the existing MLE, no new derivation needed.
- Wondering whether a summary statistic (count, mean, max) threw away useful information → check **sufficiency**: if the likelihood ratio between any two parameter values depends on the data only through that statistic, nothing was lost.

Numbers Worth Memorizing Cold

- MLE shortcuts: Binomial → k/n; Poisson → $\bar{x}$; Geometric → $1/\bar{x}$; discrete Uniform → max(data)
- MoM for discrete Uniform: $\hat{N} = 2\bar{x} - 1$
- German Tank MVUE: $\hat{N} = m + \frac{m}{k} - 1$
- $\mathbb{E}[m] = \frac{k(N+1)}{k+1}$; $\text{Var}(m) = \frac{k(N-k)(N+1)}{(k+1)^2(k+2)}$
- $\text{Var}(\hat{N}_{\text{MVUE}}) = \frac{(N-k)(N+1)}{k(k+2)}$
- MSE = Bias^2 + Var — always
- Mosteller's Locomotive Problem answers (direct MVUE applications): k=1, m=60 → $\hat{N} = 119$; k=5, m=60 → $\hat{N} = 71$
- Shrinkage/Laplace estimator: $\tilde{p} = \frac{k+1}{n+2}$